RANGEESH VENKATESAN

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OBJECTIVE

Engineer with hands-on experience with autonomous cars seeking internship in ADAS/autonomous driving applications

EDUCATION

Masters in Mechanical Engineering (Robotics)
Texas A&M University, College Station, Texas — ***GPA: 3.9/4.0*** — ***Aug 2020***

Bachelor's in Mechanical Engineering,
Indian Institute of Technology Madras (IIT-M), Chennai, India — ***GPA: 3.8/4.0*** — ***May 2020***

TECHNICAL SKILLS

Programming : C++ | Python | MATLAB | Simulink | Linux (OS) | ROS (Framework) | Julia | git | Tensorflow (GPU)
Hardware : LiDAR | RADAR | Differential GPS, RTK | IMU | CAN Bus | Wireless Communication | DSRCs

RELEVANT COURSEWORK

Optimal Control | Fundamentals of Deep Learning in CV | Machine Learning | Data Structures | Reinforcement Learning | Convex Optimization for Control System Design | Robotic Perception | Stochastic Systems

WORK EXPERIENCE

Research Intern, *TEES Center for Autonomous Vehicles & Sensor Systems*, *College Station* — **Jan 2020 - Present**
- Responsible for navigating off-road terrain with Autonomous ATV using real-time visual data from autonomous drones
- Using VFH+ Algorithm to develop Local Path Planner utilizing LiDARs for Obstacle Detection and Histogram Mapping

Controls Lead & Perception Engineer, *SAE Autodrive Challenge Team - Texas A&M* — **Mar 2019 - Present**
- Redesigned Lateral & Longitudinal Controller with superior control at higher speeds - 40+ mph (C++)
- Set up sensor fusion of LiDAR & Front Camera (Extrinsic Calibration) to identify distances of perceived objects
- Building an Extended Kalman Filter to determine speed of car utilizing GPS, IMU & Wheel Speed Sensors

Brakes & Cockpit Design Lead, *SAE Formula Student Team - IIT Madras, India* — **Aug 2015 - Aug 2018**
- Spearheaded a team of 7 and led the overall co-ordination and adherence to plan in design & manufacturing
- Introduced custom Carbon Fibre - Aluminium Sandwich Steering Wheel with 27% wt. redn. from previous iteration

RESEARCH AND ACADEMIC PROJECTS

Experimental Validation of Adaptive and Co-operative ACC for Vehicle Platooning — **Jan 2019 - Present**
Master's Thesis, Guide: Dr. Swaroop Darbha, Dr. Sivakumar Rathinam
- Formulated model equations to guarantee the boundedness and convergence of spacing errors
- Developed algorithms for filtering Radar output to find the distance of the object in front (Python)
- Validating and tuning ACC/CACC gain parameters in Simulink for multiple vehicle platooning minimizing spacing errors
- Implemented ACC on multiple vehicles and verified spacing errors using GPS | Implementing Cooperative Adaptive Cruise Control using Digital Short Relay Communications (DSRCs) for tighter spacing

Model Predictive Control of a Nonlinear Model — **Aug 2017 - Dec 2017**
Modern Control Theory - CHEM 512
- Estimated the value of the states for a Non Linear FCC Model using the Extended Kalman Filter
- Implemented MPC to control the outputs and analysed the changes in input profiles for different models of the system

Control of Automotive Systems — **Aug 2017 - Dec 2017**
- Control of an Automobile Suspension as a constrained LQR problem for both quarter and half car (roll) models
- Design and Simulation of a Collision Avoidance Controller based on the leading vehicle to maintain desired distance
- Used Constrained Optimisation Techniques to control Steering, Brakes and Throttle for Minimum time cornering

End-to-End Autonomous Driving using Imitation Learning — **Aug 2020 - Dec 2020**
Reinforcement Learning - ECEN 689
- Implemented Learning algorithms (DAgger & Safe DAgger) to enable safe driving using TORCS Simulator

AWARDS

2019	3rd Overall	SAE Autodrive Challenge Year 2 (Autonomous Car Event)
2019,2018	Scholarship	Graduate Teaching Assistant - *3 Semesters*
2018	Finalist	Raymonds Idea Challenge (Start Up)
2017	Winner	Fuel Efficiency | Formula Student Italy - Best Asian Team
2017	Award	Felicitated for Consistent Innovation by IIT Madras (SAE Formula Student)
2014	National 7th (State 1st)	National Cyber Olympiad (Out of 200,000+ Students)

ELIGIBLE TO PURSUE INTERNSHIP UNDER C.P.T.